

Linguistics as a Special Science

Philosophy of science, grammaticality, and mathematicisation

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Linguistics is not special because it can avoid mathematics

Question

Which features of language make ordinary scientific abstractions difficult?

- The object is acquired, socially sustained, historically variable, and internally structured.
- Speakers are both participants in and sources of evidence about the system.
- Linguistic categories must connect formal relations to meaningful activity.

MOLR §§2.1–2.4

Where we are

1. Philosophy of linguistics
2. Ontogeny of grammaticality
3. Fundamental questions
4. Mathematicisation

Mathematical thinking does not make linguistics a branch of mathematics

- Mathematics supplies languages of abstraction and consequence.
- Linguistics supplies the empirical objects, contrasts, and validation conditions.
- Formal precision can reveal presuppositions as well as results.
- No notation frees us from deciding what counts as evidence.

Take-away

The formalism is answerable to linguistic reality, not the reverse.

MOLR §2.1

The scientific object is produced through disciplined comparison

facts $\longrightarrow$ selected properties $\longrightarrow$ invariants $\longrightarrow$ testable relations

- Observations do not arrive already classified.
- Categories make some comparisons possible and others invisible.
- Replication requires that the selection be inspectable.

Guided analysis

Name one linguistic “fact” whose classification changes across theories.

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Children do more than repeat adult forms

"It colds my bottom." "Mummy trousers me."

- Novel forms can preserve a productive relation while violating adult lexical convention.
- Sentence–world pairings can support generalization beyond heard word–world pairings.
- Error is evidence about the child's abstraction.

Question

Which part is innovative: the category, the dependency, or the lexical choice?

MOLR §2.2

Knowing that presupposes knowing how

Ryle's distinction

Propositional descriptions of grammar do not replace the practical capacity to participate in linguistic activity.

- Speakers need not state the rules they reliably follow.
- Analysts can state rules that fail to capture speaker practice.
- Acquisition therefore concerns capacities, not merely stored propositions.

Guided analysis

Give an example of linguistic knowing-how that is difficult to report as knowing-that.

A star compresses heterogeneous judgments

- ① *She won the competition for an hour.
 - ② She won the competition for a year.
 - ③ She played the piano in a year.
 - ④ *She played the piano in an hour.
- Lexical aspect, modifier choice, context, and interpretation interact.
 - Formal grammar may be necessary without being the sole source of the contrast.

Question

What contextual repair makes each starred example better?

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Three questions constrain any theory of language

- ① Why are children able to acquire language?
- ② How are child and adult grammars related?
- ③ Why are languages diverse although dependencies appear limited?

ontogeny

continuity

diversity

Take-away

A formal description that answers only “Which strings?” is not yet a theory of these facts.

MOLR §2.3

Long-distance dependency is finite in every token, open in the system

The cat that [the dog chased ____] ran away.

- A dependency can cross material not locally selected by either endpoint.
- Each utterance is finite; the structural pattern is not bounded by a fixed depth.
- Languages differ in how the dependency is marked and restricted.

Guided analysis

List three possible overt signals of the shared element across languages.

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4. **Mathematicisation**

Using mathematics and mathematicising the object are different

Mathematics in linguistics

Use a formal technique after the empirical units and relations have been selected.

Mathematicising linguistics

Make the selection, admissibility conditions, and correspondence to reality part of the analysis.

Question

Does a statistical model mathematicise a phenomenon merely by assigning numbers?

MOLR §2.4

Third hour: diagnose an analysis

Suppose an analysis predicts every sentence in a small adult corpus.

- ① What has it described?
- ② What would acquisition require beyond this?
- ③ What continuity evidence would matter?
- ④ What typological evidence would challenge it?
- ⑤ Which assumptions must be made explicit before formalization?

Week 4 synthesis

- ① Formal work remains empirical work.
- ② Children's innovations reveal productive abstractions.
- ③ Grammaticality judgments combine several sources of admissibility.
- ④ Ontogeny, continuity, and diversity jointly constrain explanation.
- ⑤ Mathematicisation includes the bridge from symbols to linguistic reality.

Question

Quiz 3 asks what a proposed formalization establishes—and what it leaves open.